Γ: Viscosity as Structural Damping

Stokes and Navier-Stokes as limits of one equation, and the subcritical transition in pipe flow

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Self-contained manuscript beyond the algebraic theorem and equation of motion of the companion papers (Molina 2026, "Spacetime Algebra as a Theorem"; and Molina 2026, "Γ: One Equation of Motion, Three Sectors"), which this article reuses without re-deriving. Numerical verifications cited in the text are in code/ (see Appendix), published alongside this paper at https://github.com/hmolinab/papers/tree/main/gamma_fluids. Each result is marked by its status: a theorem with a complete proof, a structural correspondence (an isomorphism or algebraic relabeling with a known physical object, not a new physical theorem), a finding verified numerically or against tabulated data without a closed analytic proof, or an open frontier.

Abstract

The damping parameter γ of GSF's equation of motion ($\Gamma+\gamma\Gamma-c^2\nabla^2\Gamma+\nabla P=N$) has no fixed physical interpretation on its own: it enters through the non-conservative extension of the Lagrangian, not through the field part. This paper shows that in the fluid domain γ operationalizes exactly as $\nu=c^2(\rho)/\gamma$, the kinematic viscosity, and that this identity produces three verifiable results without tuning any new parameter. First, the Stokes equations emerge as the rigorous singular limit of the field EOM under three physically named conditions (high friction, small amplitude, short scale), each with a convergence theorem already published in the stochastic differential equations and fluid mechanics literature; and full Navier-Stokes, including the advection term, is recovered by requiring the same Galilean covariance that any continuum dynamics must respect. The identity $\nu=c^2(\rho)/\gamma$ also reproduces tabulated viscosity ratios across roughly twenty-four orders of magnitude (from mercury to Earth's mantle) with a single variable and zero fitting, and the law $\gamma\propto\rho$ is confirmed to 0.1% precision over five decades of density for air. The subcritical transition to turbulence in pipe flow (the non-modal transient growth that precedes turbulence at subcritical Reynolds numbers) is derived as a structural property of the antisymmetric sector Γ_a of the same matrix: a purely symmetric (diagonal) configuration is shown to forbid transient growth, the shear responsible for it cannot come from the potential's gradient (a Hessian is symmetric, hence normal, hence no amplification), and it does come exactly from the convective term of the material derivative. This chain reproduces the standard $G\_max\sim Re^2$ scaling with the geometric prefactor fixed from the operator's algebra, consistent in order of magnitude with the observed $Re\_c\approx 2040$ in pipe flow. The resulting framework is tested against three targets: the reduction to Stokes, the scale of real viscosity, and the mechanism of the subcritical transition, with the boundaries named where the result depends on constants taken from the literature.

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1. The parameter γ and its operationalization

1.1 Starting point (cited, not re-derived)

This paper presupposes the configuration object Γ and the equation of motion of the companion papers: $\Gamma = \Gamma\_s \oplus \Gamma\_a \in M_4(\mathbb{R})$ (the four SAIR grades over the geometric algebra G(3)) and

$$\ddot{\Gamma} + \gamma\dot{\Gamma} - c^2\nabla^2\Gamma + \nabla_\Gamma P(\Gamma,\rho) = N(t), \qquad P = \|\Gamma\|_F^2 + \mu(\rho)\det\Gamma + \beta\|\Gamma\|_F^4$$

with μ(ρ) and β fixed by the same AM-GM bound of the companion paper (zero additional free parameters). The environmental noise N(t) is paired with γ by the fluctuation-dissipation theorem.

γ does not appear in the conservative action: it enters through the Rayleigh dissipation extension, and is a coarse-graining parameter, the rate at which the unit of coherence loses memory against its environment. The fluid application considered here determines γ empirically.

1.2 The SAIR dictionary in fluids

Grade covariance alone does not fix the dictionary: the weld (Molina 2026) requires, beyond grade covariance, that S,A,I,R satisfy an explicit selection criterion whenever more than one candidate is compatible with the same grade. We use two previously established selection criteria, verified against seven domains (Newton, Schrödinger, Navier-Stokes, Maxwell, Lorentz signature, H_2O, Hopf): Gram-force consistency (Γ_s=S·A must reproduce an already-known force law independent of the domain) and the work/power criterion (among candidates of equal grade, the one that makes P=X·A≠0 generically goes to I; the one that vanishes by a pure geometric identity, not by a constraint on the flow, goes to R).

By Gram-force consistency: $S = \rho$ (density) and $A = D\mathrm{u}/Dt$ (material acceleration, not velocity) are the unique pair that make $\Gamma_s = S \cdot A = \rho\, D\mathrm{u}/Dt$ match exactly the inertial side of the Cauchy equation, the same pattern as Newton, where A is acceleration, not velocity. By the work criterion: $I = \mathrm{u}$ passes, because $\mathrm{u} \cdot (D\mathrm{u}/Dt)$ is exactly $D(|\mathrm{u}|^2/2)/Dt$, the rate of change of specific kinetic energy, generic and nonzero. The candidate that vanishes by a pure geometric identity is the Lamb vector $\mathrm{u} \times \omega$ ($\mathrm{u} \cdot (\mathrm{u} \times \omega) = 0$ always, by the triple product with a repeated vector; this is analogous to the magnetic field, which performs no work in electromagnetism), not the pressure gradient ($\mathrm{u} \cdot \nabla p$ does not vanish by identity, it only reduces to a pure divergence under the global constraint of incompressibility, §2.1 below). Since the vorticity ω that generates that vector is grade 2 and cannot occupy R directly, the grade-1 vector that produces it by wedge with u is ∇ itself:

SAIR role	Variable	Grade in $Cl_{3,0}$	Physical content
S	density ρ	0 (scalar)	inertial identity of the parcel
A	material acceleration D**u**/Dt	1 (vector)	Γ_s=S·A=ρD**u**/Dt matches the Cauchy force exactly
I	velocity u	1 (vector)	**u**·A=D(
R	∇ (operator, treated as a grade-1 generator)	1 (formal vector)	generates the field by wedge with I, without doing work, by geometric identity

The field, $\Gamma_a = I \wedge R = \mathrm{u} \wedge \nabla = \nabla \times \mathrm{u} = \omega$, is derived, not a fourth grade assigned directly: vorticity is the antisymmetric part of $\partial_j u_i$, exactly as in the companion paper for Navier-Stokes. Helicity $h = \mathrm{u} \cdot \omega$, when it appears, is the pseudoscalar invariant of Γ_a, a derived quantity, not an independent SAIR slot.

Pressure occupies no SAIR grade: it is the Lagrange multiplier of the incompressibility constraint ∇·u=0, identified via the Leray-Hodge decomposition, and enters as an effective thermodynamic forcing from the environment on the flow grades, not as an internal degree of freedom of Γ.

With this assignment, the structural law Force = S·A is exactly the left-hand side of Navier-Stokes (ρ times the material derivative of u), and the Lamb vector splits advection into a gradient part (which goes to pressure) and an irreducible part that lives in the Γ_a sector:

$$(\mathrm{u} \cdot \nabla)\mathrm{u} = \nabla\big(\tfrac{1}{2}|\mathrm{u}|^2\big) - \mathrm{u} \times \omega$$

2. Stokes as a singular limit of the EOM

2.1 The three physically named limits

Projecting the field EOM onto the velocity subsystem (with ρ fixed, as appropriate for incompressible flow), three limits reduce the EOM to the Stokes equations:

- High friction ($\varepsilon=1/(\gamma T)\to 0$): the EOM goes from second to first order in time. This is exactly the Smoluchowski-Kramers limit of high-friction Langevin dynamics, which has a published convergence theorem (Nelson, 1967; Freidlin-Wentzell): the second-order solution converges to the first-order one with error $O(\varepsilon)$.
- Small amplitude ($|v|\sim\varepsilon\to 0$): the nonlinear term $\mathrm{adj}(\Gamma)$ scales as ε^2 and becomes subdominant.
- Short scale ($L\ll c$, equivalent to low Mach number): the structural mass term, which endows the field with a screening length $\ell=c$, is negligible compared to the diffusive term $c^2\nabla^2\Gamma$. This regime is exactly the low-Mach incompressible limit, with convergence theorems available in the mathematical literature (Schochet, 2010; Lions-Masmoudi, 1998; Métivier-Schochet, 2001).

Projecting onto the active velocity component, with the pressure gradient entering as an effective thermodynamic injection from the environment, gives exactly:

$$\boxed{\partial_t v_i = \nu_{\text{kin}}\,\nabla^2 v_i - \frac{1}{\rho}\partial_i p, \qquad \nabla\cdot \mathrm{v} = 0, \qquad \nu_{\text{kin}} = \frac{c^2(\rho)}{\gamma}.}$$

Theorem (Stokes as a singular limit). Under the three coupled limits above, there exists a solution u of the Stokes equations with $\nu=c^2/\gamma$ such that $\|\Gamma-u\|\leq C\cdot\max(\varepsilon,\ \mathrm{Ma},\ L/c)$ on $[0,T]$. The existence of the limit and the form of the equation are proved, each link is a convergence theorem already published in its own domain (Smoluchowski-Kramers, low-Mach limit, Leray-Hodge); adapting those theorems from scalar/vector fields to Γ's matrix formalism, and obtaining the explicit constant C in that formalism, remains as pending work.

2.2 Full Navier-Stokes

The step from Stokes (linear) to Navier-Stokes (with advection) requires no additional postulate beyond the Galilean covariance that any continuum dynamics must respect: under a boost $x'=x-Vt$, covariance requires the time derivative to become the material derivative $D/Dt=\partial_t+(u\cdot\nabla)$ (verified symbolically that only D/Dt, not ∂_t alone, is form-invariant under the boost). Substituting that derivative into the flow slots, and using the Lamb identity to separate the gradient part (which goes to pressure) from the irreducible part (which lives in Γ_a), recovers exactly

$$\partial_t \mathrm{u} + (\mathrm{u}\cdot\nabla)\mathrm{u} = \nu\nabla^2\mathrm{u} - \tfrac{1}{\rho}\nabla p + \mathrm{f}, \qquad \nabla\cdot\mathrm{u} = 0.$$

Navier-Stokes is thus derived on the same grade-assignment postulate already used by Stokes, with no additional parameter or postulate. The remaining gap is purely algebraic: the same Clifford→$M_4(\mathbb{R})$ weld of the companion paper, not something specific to fluids.

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3. The operationalization of γ

The identity $\nu=c^2(\rho)/\gamma$ gives the first quantitative operationalization of γ in a concrete physical domain: since $\nu=c^2/\gamma$, increasing γ increases the effective viscosity, driving the flow toward the Stokes regime; low damping drives it toward the inviscid/Euler regime. The Reynolds number is the physical proxy of $1/\gamma$:

$$\mathrm{Re} = \frac{vL}{\nu_{\text{kin}}} = \frac{vL\,\gamma}{c^2(\rho)}.$$

The field stiffness $c^2(\rho)$ is taken from a scaling law of the broader GSF program (a power of ρ that only activates near the cosmological reference density), not established in either companion paper cited here; it is used in this work as an input hypothesis, not as an already-published result verifiable in those references. At any terrestrial fluid's density, that law predicts c^2 is effectively constant, so ratios of γ between fluids become pure ratios of tabulated viscosity.

4. Data without fitting

4.1 Twenty-five orders of magnitude in viscosity ratios

With c^2 constant at fluid densities, $\gamma\_A/\gamma\_B=\nu\_B/\nu\_A$ is a pure ratio of tabulated data, with no additional theoretical assumption (beyond both fluids sharing the same structural density ρ, an explicit assumption, not a data point). The kinematic viscosity values in the table below are standard engineering values at ambient temperature and pressure (Cengel and Cimbala, Fluid Mechanics: Fundamentals and Applications, 2018; CRC Handbook of Chemistry and Physics), cited here without independent re-measurement.

Fluid	ν ($\times 10^{-6}$ m^2/s)	γ relative to water
Mercury	0.12	8.4
Acetone	0.43	2.34
Water (reference)	1.004	1.000
Seawater	1.05	0.956
D_2O	1.251	0.803
Blood plasma	$\sim$1.3	$\sim$0.77
Ethanol	1.52	0.660
Glycerol (100%)	1190	8.4×10^{-4}
Glacier ice (creep)	$\sim10^{18}$	$\sim10^{-18}$
Earth's upper mantle	$\sim3\times10^{23}$	$\sim3\times10^{-24}$

The full range spans roughly twenty-four orders of magnitude (24.4, script-verified) with a single variable, a single equation, and zero fitted parameters. This comparison is independent of whether Navier-Stokes is derived from scratch: it functions as a phenomenological reference, in the same sense that Kepler's third law functioned as an empirical scaling law before Newton derived the gravity that explains it.

4.2 γ proportional to ρ, 0.1% over five decades

The kinematic viscosity of air in the continuum regime obeys $\nu\propto1/\rho$ to 0.1% precision over pressures from 10^{-3} to 10 atm. Combined with the saturation of c^2 at those densities, this implies $\gamma\propto\rho$ linearly over five decades of density, an empirical test of the structural relation between γ and density.

4.3 Absolute calibration (conditional)

The ratios above fix relative γ, not the absolute scale. Under the hypothesis that $c^2(\rho\_water)\approx c^2\_light$ (a theoretical claim, not a measurement), the implied absolute scale for water is $\gamma\approx9\times10^{22}$ s^{-1}, with a structural relaxation time of order 10^{-23} s (yoctoseconds, sub-nuclear regime, not directly accessible to femtosecond Raman/IR spectroscopy). This number is a conditional prediction under that hypothesis, not a measurement.

5. Acoustic-electromagnetic identity

In the irrotational limit ($\Gamma\_a=0$, no sources, no vorticity), the EOM reduces to the massless wave equation for the velocity potential, $\Box\varphi=0$, the same structural mechanism as the free electromagnetic photon in the companion paper. The classical acoustic-electromagnetic analogy (Bergmann, 1946), used in acoustic cloaking techniques, is not a formal analogy in this framework: both are instances of the same det=0 sector ($\gamma\approx0$, wave regime) of the same algebraic object, differing only in which physical observables occupy the slots.

6. The subcritical transition in pipe flow

6.1 Two regimes of the same equation

The Stokes reduction of §2 is leading order in $1/\gamma$ (the acceleration Γ is dropped). Retaining that term, the next-order correction gives a linearized first-order system

$$\dot{\mathrm{Y}} = \mathrm{AY}, \qquad \mathrm{A} = \begin{pmatrix} 0 & I \\ -\mathcal{L}_{\bar{\Gamma}} & -\gamma I \end{pmatrix},$$

with A structurally non-normal. Non-normal operators support transient amplification of perturbations even when all eigenvalues lie in the stable half-plane, the known mechanism of the subcritical transition in pipe flow.

6.2 The Re^2 scaling and its origin in Γ_a

For the minimal lift-up-type non-normal operator (the standard shear-amplification mechanism in hydrodynamic stability), the amplification $G(t)=\|e^{At}\|^2$ has a closed maximum proportional to Re^2, with the geometric prefactor fixed from the operator's algebra (verified numerically: log-log slope of 2.000, constant independent of Re). This recovers the standard scaling from the hydrodynamic stability literature (Reddy-Henningson, 1993; Trefethen et al., 1993) and fixes the prefactor from first algebraic principles, not by fitting.

The additional result is a structural diagnostic, verified in three steps; what exactly each step closes is stated precisely below.

1. A purely diagonal (symmetric) configuration of Γ forbids transient growth: G_max=1 for all Re. The shear coupling needed for the lift-up mechanism cannot live on the diagonal.
2. The shear cannot come from the potential's gradient $\nabla^2 P$ alone: the Hessian of any potential is symmetric (mixed derivatives commute), hence normal, hence no transient amplification is possible. This is a no-go theorem, and it correctly rules out that any generic non-symmetric perturbation of the bare EOM of Γ (without the convective term) reaches the Re^2 scaling: verified that a generic non-normal coupling, hand-built without the specific lift-up structure, gives only G_max$\sim\mathrm{Re}^1$.
3. The shear does come, exactly, from the convective term of the material derivative, linearized around a base flow profile with shear. That convective term is not foreign to this framework: it is exactly what the derivation of Navier-Stokes in §2.2 adds to the bare EOM via Galilean covariance. Linearizing that term, the resulting coupling is necessarily off-diagonal, that is, it lives in the antisymmetric sector Γ_a, and it reproduces the exact Re^2 scaling with the geometric prefactor of the previous step.

The Re^2 scaling does not come from the bare EOM of Γ with any non-symmetric perturbation (that is ruled out, point 2); it comes from the full fluid instantiation of this equation, which by Galilean covariance is Navier-Stokes with its convective term (§2.2), linearized around a base shear profile. The base profile U(y) remains an external input, the same one required by the entire standard theory of hydrodynamic stability (Orr-Sommerfeld/Squire), not something specific to this framework. Thus, Γ_a identifies the transient-growth channel: the transient growth that precedes subcritical turbulence is a direct observable of the antisymmetric sector of the configuration matrix, once the convective term that covariance requires is included.

6.3 Comparison with the observed critical Reynolds number

For Poiseuille pipe flow, this chain predicts Re_c≈2200 using a geometric constant C≈50 and a threshold perturbation amplitude $A_0\approx10^{-7}$ taken from the experimental literature (Hof et al., 2003), in reasonable agreement with the observed Re_c≈2040. What is structural here, and what depends on external constants, must be distinguished precisely:

- Structural (derived, not fitted): that the scaling is Re^2 and not another power; that this scaling strictly requires the active Γ_a sector; that the responsible shear cannot come from the gradient sector.
- Dependent on literature constants: the exact numerical value of Re_c≈2040-2200 requires the geometric constant C and the threshold amplitude A_0, both taken from published experimental measurements, not derived from the EOM.

A second, distinct regime of the same equation explains why pipe and channel differ qualitatively: channel flow transitions to turbulence via a modal mechanism (Tollmien-Schlichting) at $Re_c \approx 5772$, while pipe flow, where that mode is linearly stable for all Re, transitions via the non-modal transient-growth mechanism at $Re_c \approx 2040$. Both are the same object Γ and the same equation; what distinguishes the regimes is the Reynolds number of each geometry, not a different fluid parameter.

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7. Open problems and limitations

Frontier	Status	Note
Matrix constant C of the convergence bound to Stokes (Theorem §2.1)	open	each link of the chain of limits has a convergence theorem published in its own domain; adapting the explicit bound to Γ's matrix formalism remains pending
Mutual orthogonality of A, I, R in Γ_s (§1.2)	closed for fluids	with $A=D\mathbf{u}/Dt$, $I=\mathbf{u}$, $R=\nabla$ (verified by Gram-force consistency and the work criterion, models/calcs/brainstorming/papers/draft_a the $R=q(A,I)$ self-reference does not appear: $R=\nabla$ is independent of A and I by construction
Viscosity ratios over ~24 orders	script-verified against cited data, conditional on common ρ	code/verificacion_razones_viscosidad.py: reproduces the table within <1% except Earth's mantle (11%, an order-of-magnitude value, not a precise measurement); assumption of equal structural density between compared fluids remains explicit, not independently verified
Absolute scale of γ for water ($\gamma \approx 9\times10^{22}\ s^{-1}$)	conditional	depends on the hypothesis $c^2(\rho_{water}) \approx c^2_{light}$, an unverified theoretical claim
Homeodynamic window water/D_2O and its relation to toxicity	open frontier	calibrated from toxicity data, not derived; the causal (vs. correlational) reading is not established
$Re_c \approx 2040$ in pipe flow, exact numerical value	verified qualitatively, not first-principle	uses geometric constant and threshold amplitude from Hof et al. (2003); what is structural (Re^2 scaling, origin in Γ_a) is indeed derived
Absolute scale of ρ (needed for calibration beyond ratios)	open	referred to in the companion paper as pending program work
$c^2(\rho)$ scaling law used in §3-4	input hypothesis, not established in the companion papers cited	comes from broader GSF program material, not yet published in verifiable form

Frontier	Status	Note
Viscosity table of §4.1	no verification script yet	values cited from standard engineering sources (Cengel-Cimbala, CRC Handbook), not re-measured or reproduced in code/ yet

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8. Conclusion

γ, the one parameter of the equation of motion without a fixed interpretation from the algebraic framework alone, operationalizes precisely in fluids: $\nu=c^2/\gamma$ reproduces viscosity data across roughly twenty-four orders of magnitude without fitting anything, and $\gamma\propto\rho$ is confirmed to 0.1% over five decades. The reduction to Stokes and the derivation of full Navier-Stokes rest on convergence theorems already published in their own domains. The subcritical transition in pipe flow reduces to a closed algebraic chain: without the antisymmetric sector Γ_a there is no transient growth, that sector cannot be substituted by the potential's gradient, and it arises exactly from the convective transport term. The exact numerical value of Re_c depends on constants taken from the experimental literature; the structural result is why the scaling is Re^2 and where the mechanism that produces it lives.

References

Bergmann, P. G. (1946). The wave equation in a medium with a variable index of refraction. Journal of the Acoustical Society of America, 17(4), 329–333.

Cengel, Y. A. and Cimbala, J. M. (2018). Fluid Mechanics: Fundamentals and Applications (4th ed.). McGraw-Hill.

Freidlin, M. I. and Wentzell, A. D. (2012). Random Perturbations of Dynamical Systems (3rd ed.). Springer.

Hof, B., Juel, A., and Mullin, T. (2003). Scaling of the turbulence transition threshold in a pipe. Physical Review Letters, 91(24), 244502.

Lions, P.-L. and Masmoudi, N. (1998). Incompressible limit for a viscous compressible fluid. Journal de Mathématiques Pures et Appliquées, 77(6), 585–627.

Métivier, G. and Schochet, S. (2001). The incompressible limit of the non-isentropic Euler equations. Archive for Rational Mechanics and Analysis, 158(1), 61–90.

Moffatt, H. K. (1969). The degree of knottedness of tangled vortex lines. Journal of Fluid Mechanics, 35(1), 117–129.

Molina, H. (2026). Spacetime algebra as a theorem: deriving Cl(3,1) from the structure of a dynamical unit. DOI: 10.5281/zenodo.21184515

Molina, H. (2026). Γ: one equation of motion, three sectors: structural correspondences with Newton, Navier-Stokes, Maxwell, and Schrödinger. DOI: 10.5281/zenodo.21496578

Nelson, E. (1967). Dynamical Theories of Brownian Motion. Princeton University Press.

Reddy, S. C. and Henningson, D. S. (1993). Energy growth in viscous channel flows. Journal of Fluid Mechanics, 252, 209–238.

Schochet, S. (2010). The incompressible limit in nonlinear elasticity. In Handbook of Mathematical Fluid Dynamics, Vol. 4. Elsevier.

Trefethen, L. N., Trefethen, A. E., Reddy, S. C., and Driscoll, T. A. (1993). Hydrodynamic stability without eigenvalues. Science, 261(5121), 578–584.

Appendix. Calculation scripts

Verification scripts included in code/ alongside this paper, published at https://github.com/hmolinab/papers/tree/main/gamma_flu

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code/
  pieza2_transient_growth.py           -> G_max=Re²/C scaling (lift-up), Γ_a diagnostic, ∇²P no-go, S=∂_yU (§6)
  verificacion_razones_viscosidad.py    -> reproduces the §4.1 table from cited ν values; confirms ~24 orders of magnitude
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Requirements: numpy, scipy.

Pending implementation as an independent script (currently cited from published tables and fits, not reproduced here in code): the $\nu \propto 1/\rho$ fit for air of §4.2, and the c^2 saturation discriminating factor of §4.3.

Gamma Space Framework Program. July 2026. henrymolina@gmail.com